JF0530B Bending + Max Stress Analysis

material: low-carbon steel => E = 200GPa

↳ For a cantilever, point load applied at a free end:

$$\delta_{max} = \frac{PL^3}{3EI}$$

} we need L and I

[lets say working length 10-11mm

/ => 0.0105m

↳ $I_x = 3.94\,mm^4$

← — — — — at most, applied load is 700N

$$\delta_{max} = \frac{P(0.0105)^3}{3(200\cdot10^9)\cdot3.98\cdot10^{-12}} =$$

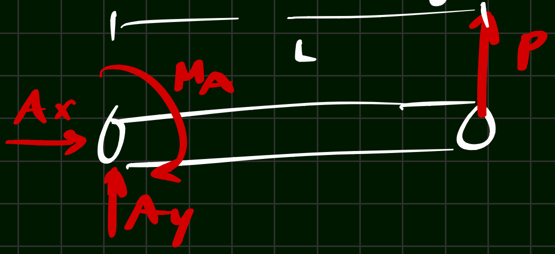
$$\boxed{\delta_{max} = 0.34\,mm}$$

← confirm in Ansys

↳ For this cylinder, max-stress is likely to ⓐ be at the fixed end, ⓑ be at the cross-section that is reduced

↳ I predict the scenario ⓐ

$$\rightarrow \sigma_{max} = \frac{ML}{I} \quad \begin{matrix} \} \; 0.0105 \\ \} \; I_x = \frac{\pi(0.003)^4}{4} \end{matrix}$$



} static Analysis

$$\Sigma F_y = A_y = 0 \qquad M_A = -PL$$
$$A_x = 0$$

$$\sigma_{max} = \frac{PLr}{I} = \frac{700 \cdot 0.0105(0.003)}{\frac{\pi(0.003)^4}{4}} \Rightarrow$$

$$\boxed{\sigma_{max} = 346.6 \text{ MPa}}$$

→ verify in Ansys

AISI $\sigma_{critical}$ is 370 MPa, so we're cutting it close

↳ 700 N is roughly 157 lb grip strength, so not only does this have to be applied on the small - small surface area ($P = \frac{700}{0.000781} = 89.6$ kPa) but, this has to be an elite human